(-3) multiplied with another negative integer -5? To understand this, let us observe the following pattern.

$$(-5)\times 3 = -15$$
$$(-5)\times 2 = -10$$
$$(-5)\times 1 = -5$$
$$(-5)\times 0 = 0$$
$$(-5)\times(-1) = +5$$
$$(-5)\times(-2) = +10$$
$$(-5)\times(-3) = +15$$

Note that how in each step the number increase by 5 from -15 to -10, from -10 to -5, from -5 to 0. Therefore, the next number in the pattern will be +5 only and not -5. Similarly $(-5)\times(-2)$ is a positive integer 10. So, the product of two negative integers is always a positive integer.

Try these

1. Find the product of the following

(i) $(-20)\times(-45) =$ _____ (ii) $(-9)\times(-8) =$ _____

(iii) $(-30)\times 40\times(-1) =$ _____ (iv) $(+50)\times 2\times(-10) =$ _____

2. Complete the following table by multiplying the integers in the corresponding row and column headers.

X	-3	-2	-1	0	1	2	3
-3							
-2							
-1							
0							
1							
2							
3							

3. Which of the following is incorrect?

(i) $(-55)\times(-22)\times(-33) < 0$

(ii) $(-1521)\times 2511 < 0$

(iii) $2512 - 1252 < 0$

(iv) $(+1981)\times(+2000) < 0$

Think

Can you express the product 15×16 as sum or difference of integers?
Yes, here are 4 ways to compute $15\times 16 = +240$.

(i) $15\times 16 = (10+5)\times(10+6) = 100+60+50+30 = 240$

(ii) $15\times 16 = (20-5)\times(10+6) = 200+120+(-50)+(-30) = 240$

(iii) $15\times 16 = (10+5)\times(20-4) = 200+(-40)+100+(-20) = 240$

(iv) $15\times 16 = (20-5)\times(20-4) = 400+(-80)+(-100)+\boxed{20} = 240$

From the above pattern one can determine the product of two positive integers, two negative integers, one positive integer with one negative integer.

Example 1.17 Find the value of :

(i) $(-35) \times (-11)$ (ii) $96 \times (-20)$ (iii) $(-5) \times 12$

(iv) 15×5 (v) 999×0

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